

Vibrations: Free and forced vibration of single degree of freedom systems, effect of damping; vibration isolation; resonance; critical speeds of shafts.

Machine Design: Design for static and dynamic loading; failure theories; fatigue strength and the SN diagram; principles of the design of machine elements such as bolted, riveted and welded joints; shafts, gears, rolling and sliding contact bearings, brakes and clutches, springs.

Section 2: Fluid Mechanics and Thermal Sciences

Fluid Mechanics: Fluid properties; fluid statics, forces on submerged bodies, stability of floating bodies; control-volume analysis of mass, momentum and energy; fluid acceleration; differential equations of continuity and momentum; Bernoulli's equation; dimensional analysis; viscous flow of incompressible fluids, boundary layer, elementary turbulent flow, flow through pipes, head losses in pipes, bends and fittings; basics of compressible fluid flow.

Heat-Transfer: Modes of heat transfer; one dimensional heat conduction, resistance concept and electrical analogy, heat transfer through fins; unsteady heat conduction, lumped parameter system, Heisler's charts; thermal boundary layer, dimensionless parameters in free and forced convective heat transfer, heat transfer correlations for flow over flat plates and through pipes, effect of turbulence; heat exchanger performance, LMTD and NTU methods; radiative heat transfer, Stefan-Boltzmann law, Wien's displacement law, black and grey surfaces, view factors, radiation network analysis

Thermodynamics: Thermodynamic systems and processes; properties of pure substances, behavior of ideal and real gases; zeroth and first laws of thermodynamics, calculation of work and heat in various processes; second law of thermodynamics; thermodynamic property charts and tables, availability and irreversibility; thermodynamic relations.

Applications: Power Engineering: Air and gas compressors; vapour and gas power cycles, concepts of regeneration and reheat. I.C.

Engines: Air-standard Otto, Diesel and dual cycles. Refrigeration and air-conditioning: Vapour and gas refrigeration and heat pump cycles; properties of moist air, psychrometric chart, basic psychrometric processes. Turbo machinery: Impulse and reaction principles, velocity diagrams, Pelton-wheel, Francis and Kaplan turbines; steam and gas turbines.

Section 3: Materials and Manufacturing

Engineering Materials: Structure and properties of engineering materials, phase diagrams, heat treatment, stress-strain diagrams for engineering materials.

Casting, Forming and Joining Processes: Different types of castings, design of patterns, moulds and cores; solidification and cooling; riser and gating design. Plastic deformation and yield criteria; fundamentals of hot and cold working processes; load estimation for bulk (forging, rolling, extrusion, drawing) and sheet (shearing, deep drawing, bending) metal forming processes; principles of powder metallurgy. Principles of welding, brazing, soldering and adhesive bonding.

Machining and Machine Tool Operations: Mechanics of machining; basic machine tools; single and multi-point cutting tools, tool geometry and materials, tool life and wear; economics of machining; principles of non-traditional machining processes; principles of work holding, jigs and fixtures; abrasive machining processes; NC/CNC machines and CNC programming.

Section 4: Industrial Engineering

Metrology and Inspection: Limits, fits and tolerances; linear and angular measurements; comparators; interferometry; form and finish measurement; alignment and testing methods; tolerance analysis in manufacturing and assembly; concepts of coordinate-measuring machine (CMM).

Computer Integrated Manufacturing: Basic concepts of CAD/CAM and their integration tools; additive manufacturing. Robotics and Mechatronics.

Production Planning and Control: Forecasting models, aggregate production planning, scheduling, materials requirement planning; lean manufacturing.

Inventory Control: Deterministic models; safety stock inventory control systems.

Operations Research: Linear programming, simplex method, transportation, assignment, network flow models, simple queuing models, PERT and CPM.

13. USME

Discipline: Management

Managerial themes such as Management Principles, Information Technology Management, Business policy and Strategic Management, Marketing Management, Distribution and Retail Management, Brand and product management, Organizational Behavior and Development, Human Resources Management, Performance and talent management, Corporate Governance and Ethics, Public Policy and Governance, Accounting and Finance, Portfolio Management, Mergers and Acquisition, Corporate Restructuring, Knowledge Management and Practices, Supply chain Management, Business Research Methods, Business Statistics.

Operations management, vendor management, Services marketing and service operations, CRM, Business Analytics, predictive techniques, marketing financial and HR analytics, supply chain analytics, optimization techniques, big data analytics. Entrepreneurship, innovation management, managing technology. Managerial Economics, forecasting, inventory management, various types of costing approaches, total quality management.

Discipline: Innovation, Entrepreneurship and Venture Development

Unit 1: Comprehension Passages from success stories in Entrepreneurship and startup Leadership. Current affairs related to start-ups.

Unit 2: Entrepreneurship, Social entrepreneurship, Intrapreneurship, Entrepreneurial Characteristics, Entrepreneur v/s Manager, Entrepreneurial Motivation.

Unit 3 : Innovation Management, Diffusion of Innovation, Innovation Cycle, New Product Development, Design Thinking.

Unit 4 : Startup, Stages of Startup, Ideation, Pre-startup, Startup and Scaling up Stage, Start Up India Program, Start Up concept and its support to start Ups, Startup Financing, Seed Funding, Angel investors, venture capital funding, government financial support for startups, MSME-definition and its types.

Unit 5: Government Schemes & Policy for Startups and MSMEs, ATAL Innovation Mission (AIM), ATAL Tinkering Lab(ATL), Self Employment & Talent Utilization (SETU) scheme, SFURTI Scheme, Intellectual Property Rights (IPR), Start Up Intellectual Property Protection (SIPP), Trademark, Copyright, Patents, Type of Companies in India and their Characteristics, Formation Legalities of the Company.

14. Centre of Excellence for the Science of Happiness (CESH)

Science and practice of happiness, Measurement of Happiness and well being, Happiness at work and daily living, consumer happiness, building resilience, Gross National Happiness, Traditional knowledge and Values, Theories of emotions, Happiness and prosperity, Harmony in family and society, Human values, Science and art behind meditation, Power of Self discipline, Peaceful conversation, Brain and heart connection, Healing from the heart, Managing with heart.

15. Department of Geospatial Science and Technology (DGST)

Engineering Mathematics:- Surveying measurements, Accuracy, Precision, Most probable value, Errors and their adjustments, Regression analysis, Correlation coefficient, Least square adjustment, Statistical significant value, Chi square test. Remote Sensing: Basic concept, Electromagnetic spectrum, Spectral signature, Resolutions- Spectral. Spatial, Temporal and Radiometric, Platforms and Sensors, Remote Sensing Data Products - PAN, Multispectral, Microwave, Thermal, Hyperspectral, Visual and digital interpretation methods GNSS: Principle used, Components of GNSS, Data collection methods, DGPS, Errors in observations and corrections. GIS: